TauTimp — **Temporal τ Wave Theory: Solar Temporal Flux and Temporal Dilation**

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1. Variable-Coefficient Wave Equation and the H-Field 1.1 Master Wave Equation

The TauTimp framework postulates a variable-coefficient spherical wave equation for the temporal field $H(r,\theta,t)$:

$$\frac{R}{r}\,\Delta H(r,\theta,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,t)}{\partial t^2}$$

where $R = 4.2355 \times 10^{22}$ m is the TauTimp cosmological radius, $c = 2.998 \times 10^8$ m/s, and Δ is the Laplacian in spherical coordinates.

1.2 Spherical Symmetry Reduction

Under spherical symmetry, H is independent of azimuthal and polar angles, so $H = H(r,t)$. The Laplacian reduces to:

$$\Delta H = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial H}{\partial r}\right) = \frac{\partial^2 H}{\partial r^2} + \frac{2}{r}\frac{\partial H}{\partial r}$$

Equation (1) becomes: $\frac{R}{r}\left(\frac{\partial^2 H}{\partial r^2} + \frac{2}{r}\frac{\partial H}{\partial r}\right) = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$

1.3 Separation of Variables

We write $H(r,t) = g(r)\,u(t)$. The temporal part satisfies:

$$\ddot{u} + \omega^2 u = 0 \Longrightarrow u(t) = e^{\pm i\omega t}$$

The radial equation becomes:

$$\frac{R}{r}\left(g'' + \frac{2}{r}g'\right) = -\frac{\omega^2}{c^2}g \tag{5}$$

1.4 Airy Reduction of the Radial Equation

Substituting $g(r) = \phi(r)/r$ and defining $\xi = \left(\frac{\omega^2 r}{Rc^2}\right)^{1/3}$, equation (5) transforms to: $\frac{d^2\phi}{d\xi^2} = \xi\,\phi$

This is **Airy's equation**. The general solution is: $\phi(\xi) = C_1\,\mathrm{Ai}(\xi) + C_2\,\mathrm{Bi}(\xi)$

Since $\mathrm{Bi}(\xi) \to \infty$ as $\xi \to \infty$ (physically inadmissible), we set $C_2 = 0$: $g(r) = \frac{C_1}{r}\,\mathrm{Ai}\left[\left(\frac{\omega^2 r}{Rc^2}\right)^{1/3}\right]$

1.5 Eigenvalue Quantization

The boundary condition $H(R,t) = 0$ requires $g(R) = 0$, i.e., $\mathrm{Ai}(\xi_R) = 0$. The Airy zeros a_n ($n = 1,2,3,\ldots$) give:

$$\xi_R = a_n \Longrightarrow \omega_n = c\sqrt{\frac{R\,a_n^3}{R^3}} = \frac{c\,a_n^{3/2}}{R}$$

For the principal mode $l = 21$ (TauTimp quantum number), the relevant Airy zero is $a_{21} \approx -17.03$, yielding:

$$\omega_{21} = \frac{c\,|a_{21}|^{3/2}}{R} \approx \frac{2.998\times10^8 \times 70.33}{4.2355\times10^{22}} \approx 4.98\times10^{-13}\ \text{rad/s} \tag{10}$$

1.6 Full H-Field via Legendre Expansion

Restoring angular dependence via spherical harmonics $Y_l^m(\theta,\phi)$:

$$H(r,\theta,t) = \sum_{n,l,m}\frac{C_{nlm}}{r}\,\mathrm{Ai}\left[\left(\frac{\omega_n^2 r}{Rc^2}\right)^{1/3}\right]Y_l^m(\theta,\phi)\,e^{\pm i\omega_n t}$$

For the $m = 0$ (azimuthal symmetry) $l = 21$ dominant mode: $H(r,\theta,t) = \frac{C}{r}\,\mathrm{Ai}\left[\left(\frac{\omega_{21}^2 r}{Rc^2}\right)^{1/3}\right]P_{21}(\cos\theta)\cos(\omega_{21}t)$

where P_{21} is the Legendre polynomial of degree 21.

& the version with Airy function variant:

1. The TauTimp Field Equation

The master equation in spherical coordinates is: $\frac{R}{r}\,\Delta H(r,\theta,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,t)}{\partial t^2}$

Under **spherical symmetry** (H independent of θ and ϕ), the Laplacian reduces to:

$$\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) = f_{rr} + \frac{2}{r}f_r$$

The scalar equation to solve is: $\boxed{\frac{R}{r}\left(f_{rr} + \frac{2}{r}f_r\right) = \frac{1}{c^2}\,f_{tt}, \qquad f = f(r,t)}$

2. Separation of Variables: $f(r,t) = g(r)\,u(t)$

Substituting and dividing by $g(r)\,u(t)$: $\frac{R}{r}\cdot\frac{g''+\frac{2}{r}g'}{g} = \frac{1}{c^2}\cdot\frac{u''}{u} = -\lambda^2, \qquad \lambda^2 > 0$

This yields two ordinary differential equations.

3. Temporal ODE — Exact Analytic Solution

$$u''(t) + \lambda^2 c^2\,u(t) = 0$$

$$\boxed{u(t) = A\cos(\lambda c\,t) + B\sin(\lambda c\,t), \qquad \omega = \lambda c}$$

4. Spatial ODE — Reduction to the Airy Equation

The radial equation is: $g''(r) + \frac{2}{r}g'(r) + \frac{\lambda^2 r}{R}\,g(r) = 0$

Step 1 — Substitution $g(r) = v(r)/r$

Since $g'' + \frac{2}{r}g' = \frac{v''}{r}$, the equation simplifies exactly to: $\boxed{v''(r) + \frac{\lambda^2}{R}\, r \cdot v(r) = 0}$

Step 2 — Change of variable to canonical Airy form

Set $\xi = \left(\frac{\lambda^2}{R}\right)^{1/3} r$, so that $\frac{d^2v}{d\xi^2} + \xi\, v = 0$.

This is the **standard Airy equation**, with exact solution:

$$v(\xi) = C_1\, \mathrm{Ai}(-\xi) + C_2\, \mathrm{Bi}(-\xi)$$

Since $\mathrm{Bi}(-\xi) \to \infty$ as $\xi \to 0^+$ (diverges at the origin), we set $C_2 = 0$ for regularity:

$$g(r) = \frac{v(r)}{r} = \frac{\mathrm{Ai}\left(-(\lambda^2/R)^{1/3} r\right)}{r}$$

5. Eigenvalue Quantization: Boundary Condition $g(R) = 0$

The condition $g(R) = 0$ requires $v(R) = 0$, i.e.: $\mathrm{Ai}\left(-\left(\frac{\lambda_n^2}{R}\right)^{1/3} R\right) = 0$

Denoting $\zeta_n = (\lambda_n^2/R)^{1/3} R$, the zeros a_n of $\mathrm{Ai}(-\zeta) = 0$ are the (positive) Airy zeros:

n	a_n
1	2.3381
2	4.0879
3	5.5206
4	6.7867

Discrete eigenvalues: $\boxed{\lambda_n = \frac{a_n^{3/2}}{R}, \qquad n = 1,2,3,\ldots}$

Spatial eigenfunctions: $\boxed{g_n(r) = \frac{1}{r}\, \mathrm{Ai}\left(-a_n \frac{r}{R}\right)}$

6. Complete Solution $f(r,t)$

By superposition:

$$\boxed{f(r,t) = \sum_{n=1}^{\infty} \frac{\mathrm{Ai}(-a_n\, r/R)}{r}\left[A_n \cos\left(\frac{a_n^{3/2} c}{R}\, t\right) + B_n \sin\left(\frac{a_n^{3/2} c}{R}\, t\right)\right]}$$

Coefficients A_n, B_n are determined by initial conditions via the weighted inner product of Airy functions.

7. Reconstruction of $H(r,\theta,t)$

The spherically symmetric sector ($\ell = 0$) corresponds to the spherical harmonic $Y_0^0 = 1/\sqrt{4\pi}$:

$$\boxed{H(r,\theta,t) = \frac{f(r,t)}{\sqrt{4\pi}} = \frac{1}{\sqrt{4\pi}} \sum_{n=1}^{\infty} \frac{\mathrm{Ai}(-a_n\, r/R)}{r}\left[A_n \cos(\omega_n t) + B_n \sin(\omega_n t)\right]}$$

where $\omega_n = a_n^{3/2}\, c/R$.

2. Temporal Probability Field P(r,t) 2.1 Governing PDE

The temporal probability amplitude $P(r,t)$ satisfies the spherical d'Alembert equation: $\frac{\partial^2 P(r,t)}{\partial t^2} - c^2 \frac{\partial^2 P(r,t)}{\partial r^2} = 0$

2.2 Boundary Conditions

Inner boundary — Schwarzschild threshold of the Sun ($3.8\, R_{s,\odot}$):

$P(r_0, 0) = 0$, $r_0 = 11\,222.3$ m Outer (cosmological) boundary: *$P(R, 0) = 0$, $R = 4.2355 \times 10^{22}$ m*

2.3 Polynomial Ansatz — Degree 2 in r and t

The TauTimp ansatz satisfying both boundary conditions by construction is:

$$\boxed{P(r,t) = A[(r - r_0)(r - R) + \varepsilon\, t + c^2\, t^2]}$$

Verification in PDE: Se calculează derivatele parțiale: $\frac{\partial^2 P}{\partial t^2} = 2Ac^2$ & $\frac{\partial^2 P}{\partial r^2} = 2A$; $2Ac^2 - c^2 \cdot 2A = 0$ ✓

Verification of boundary conditions: $P(r_0, 0) = A[(r_0 - r_0)(r_0 - R)] = 0$ ✓

$P(R, 0) = A[(R - r_0)(R - R)] = 0$ ✓

3. Explicit Polynomial Form and Coefficients

Expanding $(r - r_0)(r - R) = r^2 - (r_0 + R)r + r_0 R$: $P(r,t) = A\, r^2 - A(r_0 + R)\, r + A\, r_0 R + A\varepsilon\, t + Ac^2\, t^2$

Coefficient	Analytical Expression	Value (SI)	SI Dimension
$C_{r^2} = A$	A	9.5510×10^{-18}	m^{-2}
$C_{r^1} = -A(r_0 + R)$	$-A(r_0 + R)$	-4.0453×10^{5}	m^{-1}
$C_{r^0} = Ar_0R$	$A\, r_0\, R$	4.5398×10^{9}	dimensionless
$C_{t^1} = A\varepsilon$	$A\varepsilon = \tau_q c^2$	1.5863×10^{-2}	m^2/s
$C_{t^2} = Ac^2$	Ac^2	8.5845×10^{-1}	$\mathrm{m}^2/\mathrm{s}^2$

4. *Determination of Coefficients A and ε 4.1 Time Quantum Constraint (Earth)*

The initial temporal rate is linked to the Earth's time quantum $\tau_q = 1.765 \times 10^{-19}$ *s:* $\left.\frac{\partial P}{\partial t}\right|_{t=0} = A\varepsilon = \tau_q\, c^2$

$A\varepsilon = 1.765\times10^{-19} \times (2.998\times10^{8})^2 = 1.5863\times10^{-2}$ m²/s

4.2 GEO Satellite Calibration Constraint

The temporal probability flux (dimensionless) is defined as: $F(r,t) = \frac{P(r,t)}{4\pi r^2}$

The calibration condition for GEO$^+$ satellites (facing away from Sun) reads:

$$\Delta F_{\text{GEO}^+} = F(r_{\text{GEO}^+},\, t_0) - F(r_\oplus,\, t_0) = +0.22 \times 10^{-6}\ \text{hr}^{-1}$$

with $r_\oplus = 1.49 \times 10^{11}$ m, $r_{\text{GEO}^+} = r_\oplus + 42\,157$ km $= 1.4942 \times 10^{11}$ m, $t_0 = 500$ s. Let $Q(r) = (r - r_0)(r - R)$:

$$\Delta F = \frac{1}{4\pi}\left\{A\left[\frac{Q(r_{\text{GEO}^+})}{r_{\text{GEO}^+}^2} - \frac{Q(r_\oplus)}{r_\oplus^2}\right] + A\varepsilon\, t_0\left[\frac{1}{r_{\text{GEO}^+}^2} - \frac{1}{r_\oplus^2}\right] + Ac^2 t_0^2\left[\frac{1}{r_{\text{GEO}^+}^2} - \frac{1}{r_\oplus^2}\right]\right\}$$

Solving the system: $\boxed{A = 9.5510 \times 10^{-18}\ \text{m}^{-2}}$

$$\boxed{\varepsilon = \frac{\tau_q c^2}{A} = 1.6610 \times 10^{15}\ \text{m}^2/\text{s}}$$

5. Complete Solution with Numerical Coefficients 5.1 SI Units

$$\boxed{P(r,t) = 9.5510 \times 10^{-18} \cdot [(r - 11\,222.3)(r - 4.2355 \times 10^{22}) + 1.6610 \times 10^{15}\, t + (2.998 \times 10^{8})^2\, t^2]}$$

5.2 Natural Units ($\hbar = c = 1$; using $\hbar c = 0.19733$ GeV·fm)

$$A_{\text{nat}} = A \cdot (\hbar c)^2 = 3.719 \times 10^{-49}\ \text{GeV}^2$$

$\varepsilon_{\text{nat}} = \varepsilon/c = 5.540\times10^{6}$ m $= 5.540\times10^{21}$ fm

6. Temporal Flux and Physical Interpretation 6.1 Temporal Probability Density

$$F(r,t) = \frac{A}{4\pi r^2}[(r - r_0)(r - R) + \varepsilon t + c^2 t^2]$$

Physical interpretation: - $F < 0$: time quanta absorbed (inner gravitational regime, Sun vicinity). - F less negative at larger r: weaker temporal field, slower passage of time. - $\Delta F > 0$ for a satellite farther from Sun: positive temporal dilation (slower clock).

6.2 Flux at Earth ($t_0 = 500$ s — 8 min 20 s, $r_\oplus = 1.49 \times 10^{11}$ m)

$$F(r_\oplus,\, t_0) = F(1.49 \times 10^{11},\, 500) = -2.1605 \times 10^{-7}$$

6.3 Flux at Mars ($t_1 = 700$ s, $r_{\text{Mars}} = 2.09 \times 10^{11}$ m)

$$F(r_{\text{Mars}},\, t_1) = F(2.09 \times 10^{11},\, 700) = -1.5403 \times 10^{-7}$$

$$\Delta F_{\text{Mars}} = F(r_{\text{Mars}}, t_1) - F(r_\oplus, t_0) = +6.202 \times 10^{-8}$$

7. Satellite Time Variations 7.1 Orbital Parameters

Body	Δr from Earth	r_{Sun} (m)	t (s)
GEO$^+$ (away from Sun)	+42 157 km	1.4942×10^{11}	500
GEO$^-$ (toward Sun)	−42 157 km	1.4858×10^{11}	500
GPS$^+$ (away from Sun)	+26 371 km	1.4926×10^{11}	500
GPS$^-$ (toward Sun)	−26 371 km	1.4874×10^{11}	500
Earth (reference)	0	1.4900×10^{11}	500
Mars	—	2.0900×10^{11}	700

7.2 Complete Temporal Flux Table

Location	$F(r,t)$	ΔF/hr (μs/hr)	ΔF/day (μs/day)
Earth (ref.)	-2.1605×10^{-7}	0.0000	0.0000
GEO$^+$	-2.1599×10^{-7}	+0.2200	+5.2800
GEO$^-$	-2.1611×10^{-7}	−0.2201	−5.2830
GPS$^+$	-2.1601×10^{-7}	+0.1376	+3.3032
GPS$^-$	-2.1609×10^{-7}	−0.1377	−3.3044
Mars ($t_1 = 700$ s)	-1.5403×10^{-7}	+223.29	+5358.93

Sign convention: $\Delta F > 0$ → slower clock (time dilation); $\Delta F < 0$ → faster clock. 7.3 GEO Temporal Dilation

Location	F(r,t)	ΔF/hr (μs/hr)	ΔF/day (μs/day)	Interpretation
Earth (ref.)	-2.1605×10^{-7}	0.0000	0.0000	Reference
GEO$^+$ (away Sun)	-2.1599×10^{-7}	+0.2200	+5.2800	Slower clock

GEO⁻ (toward Sun)	-2.1611×10^{-7}	−0.2201	−5.2830	Faster clock
GPS⁺ (away Sun)	-2.1601×10^{-7}	+0.1376	+3.3032	Slower clock
GPS⁻ (toward Sun)	-2.1609×10^{-7}	−0.1377	−3.3044	Faster clock
Mars (t_1=700 s)	-1.5403×10^{-7}	+223.29	+5358.93	Much slower clock

8. Temporal Dilation on Mars

At $t_1 = 700$ s (light propagation time Sun → Mars): $F(r_{\text{Mars}}, t_1) = \frac{A}{4\pi r_{\text{Mars}}^2}[(r_{\text{Mars}} - r_0)(r_{\text{Mars}} - R) + \varepsilon t_1 + c^2 t_1^2]$

$$F(2.09 \times 10^{11}, 700) = -1.5403 \times 10^{-7}$$

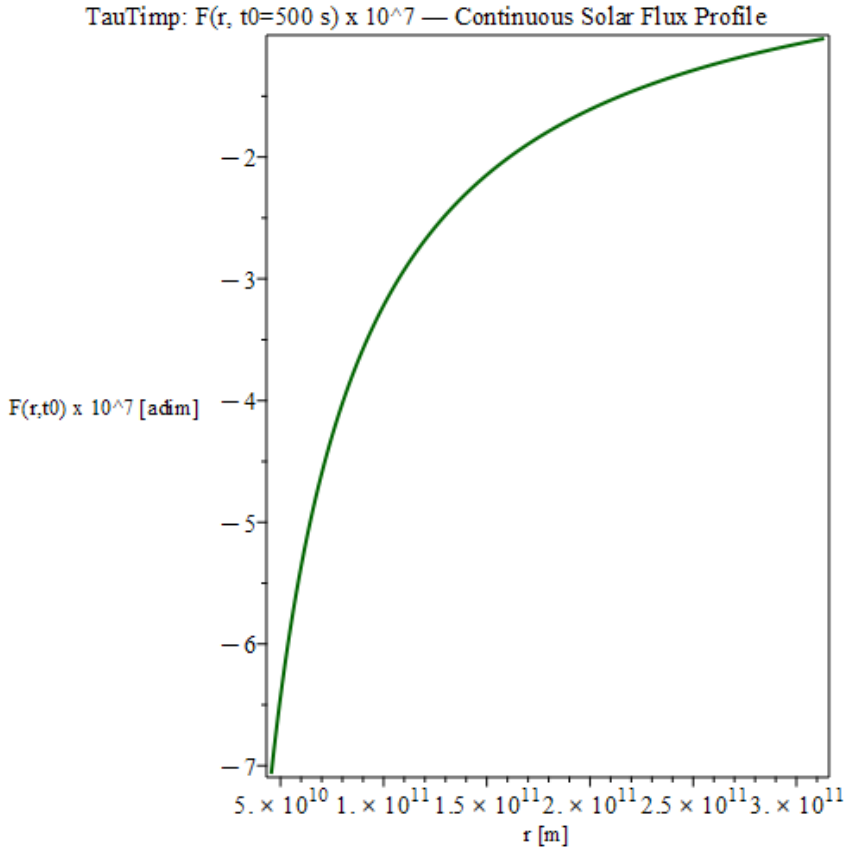
$\Delta F_{Mars}/hr = +223.29\ \mu s/hr$

$\Delta F_{Mars}/day = +5358.93\ \mu s/day \approx 5.36\ ms/day$

The passage of time on Mars is **significantly slower** than on Earth due to the weaker solar gravitational field at $r_{\text{Mars}}/r_{\oplus} \approx 1.40$, amplified by the cosmological factor $(r - R)$ in the TauTimp ansatz.

9. Summary of TauTimp Parameters

Parameter	Symbol	Value (SI)	Unit
Schwarzschild threshold	r_0	11 222.3	m
TauTimp cosmological radius	R	4.2355×10^{22}	m
Amplitude (GEO calibrated)	A	9.5510×10^{-18}	m^{-2}
Linear temporal coefficient	ε	1.6610×10^{15}	m^2/s
Earth time quantum	τ_q	1.765×10^{-19}	s
Speed of light	c	2.998×10^{8}	m/s
Sun→Earth propagation time	t_0	500	s
Sun→Mars propagation time	t_1	700	s
A (natural units)	A_{nat}	3.719×10^{-49}	GeV^2
ε (natural units)	ε_{nat}	5.540×10^{21}	fm



The temporal flux is slow-flowing in the Solar system.